Analysis and Recommendations for Boosting Sales and Loyalty at Dibs

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Introduction:

Dibs is a rapidly expanding retail company that sells a wide variety of goods, such as shoes, clothes, food, drinks, and souvenirs etc. Although its rapid expansion, the organization needs help in expanding revenue and strengthening support. Dibs has gathered a lot of information on past purchases made by customers at different malls to address these problems, but it lacks the understanding to make the most of this information.
This report aims to examine customer purchase data to identify trends and patterns that can guide Dibs' marketing and sales strategies. The business can create focused marketing targeted at boosting sales and cultivating customer loyalty by getting a deeper understanding of client behavior.

Data Cleaning:

A vital phase in getting the dataset ready for analysis is data cleansing. The accuracy, consistency, and suitability of the data for additional analysis are ensured by this procedure. The data was cleaned by doing the subsequent actions, which fixed problems like duplicates, outliers, missing values, and inconsistent data types. Steps are below.

1. **File Import**

- The data cleaning procedure was started when the dataset was successfully imported into the analysis environment.

2. **Date Formatting**

- The invoice dates were standardized into a mixed format for clarity. Missing values in the date column were corrected, and the month and year were extracted from the invoice dates to facilitate time-based analysis.

3. **Managing Missing Values**

- I identified three missing values using the isnull() and isna() functions—two in the price column and one in the payment method column.

4. **Data Correction**

- Inconsistent textual data was corrected. For example, variations such as "male," "female," and "mal" were found, and "mal" was corrected to "male" for consistency. Any incorrect values in the dataset were also rectified.

5. **Dropping Unnecessary Data**

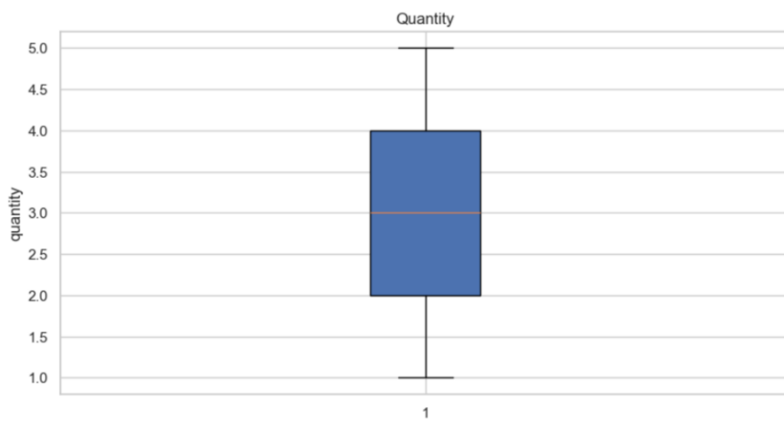
- The rows with missing values were removed, and duplicate records were eliminated to ensure that each transaction was unique.

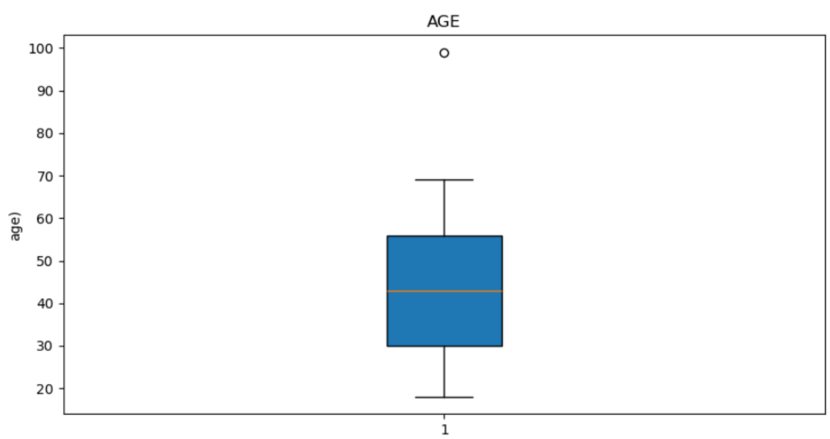
6. **Histogram Analysis**

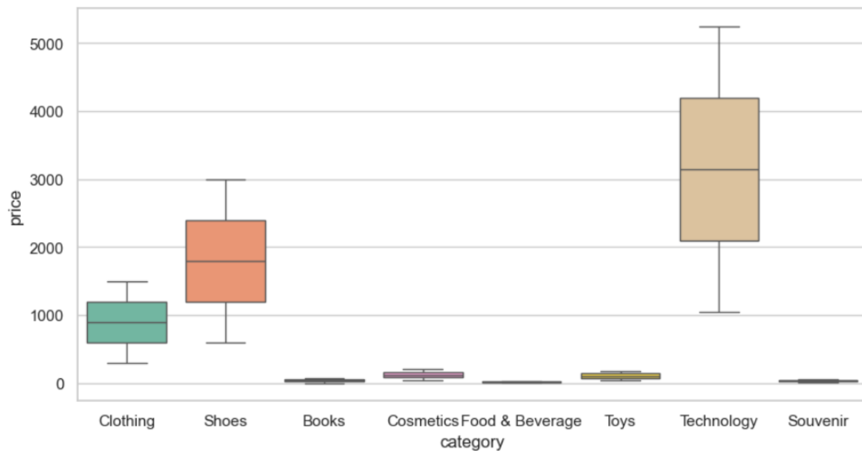
- Histograms were generated to visualize the distribution of key variables, which helped in identifying any irregularities and understanding the overall data distribution.

7. **Outlier Detection and Handling**

- Box plots for variables such as age, quantity, and price per category were analyzed to detect outliers. The analysis showed no significant outliers So the data was within an acceptable range.







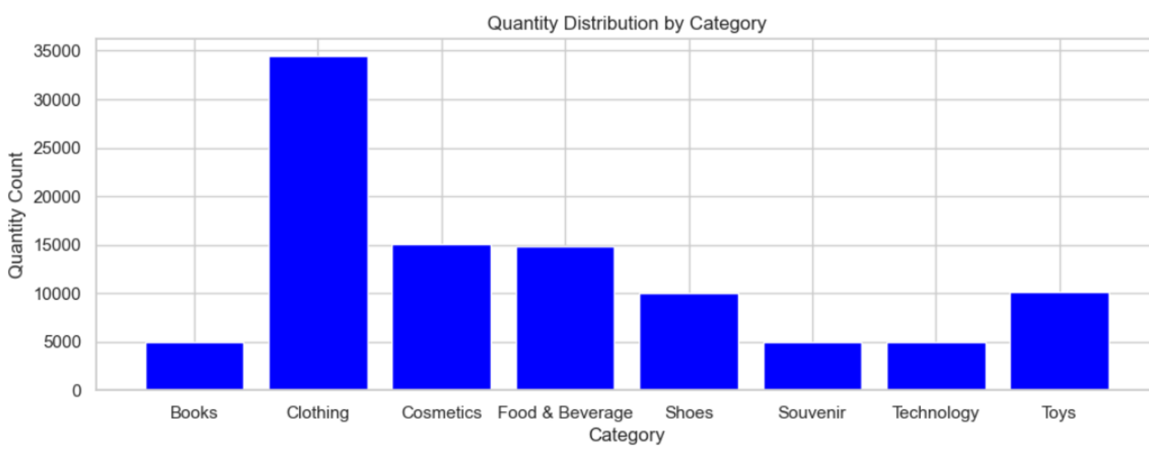
Analysis:

Customer Segmentation: Based on the bar chart, the shopping mall should consider targeting the following customer segments/categories for its marketing campaigns:

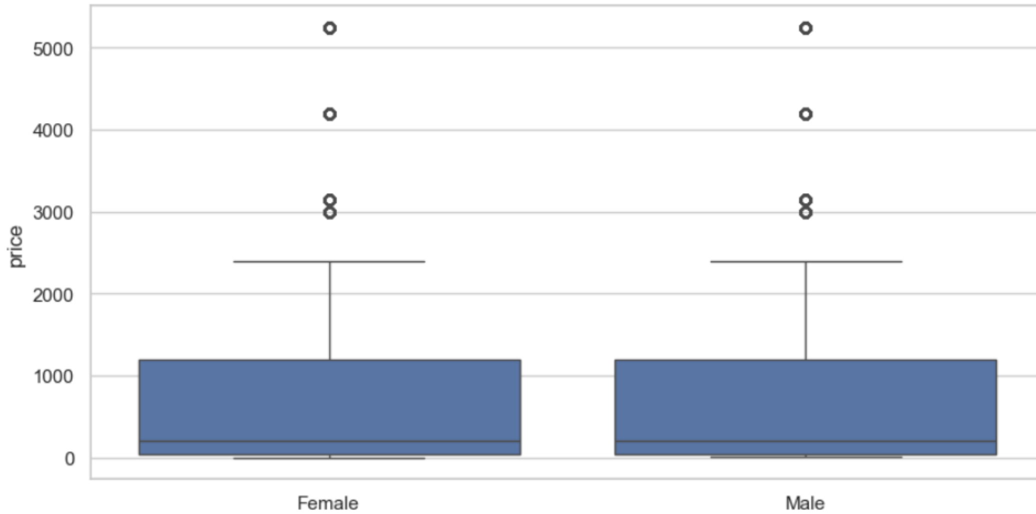
1. Clothing: This category has the highest quantity count, indicating a significant demand for clothing items.

2. Cosmetics: While it's not as high as clothing, the cosmetics category still shows a substantial interest among shoppers.

3. Food & Beverage: This category represents a consistent demand, likely due to the need for food and drink options within the shopping mall.

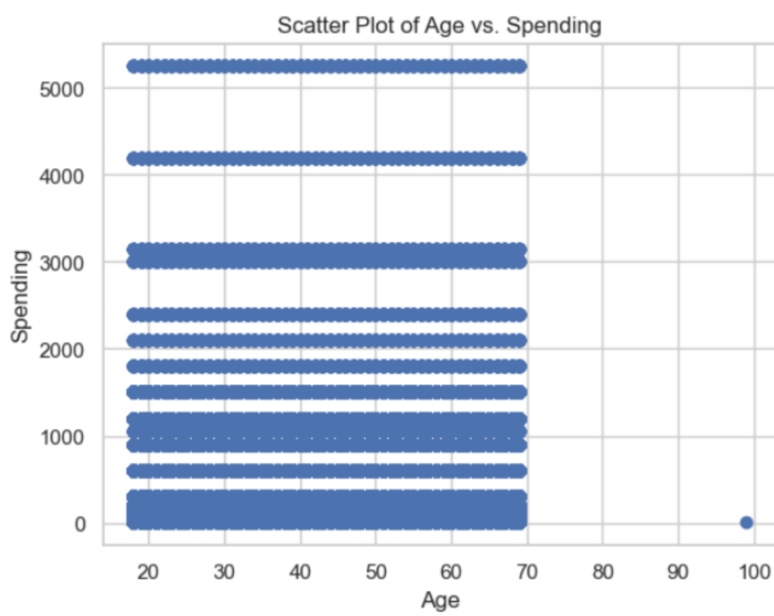


Gender-Based Spending Behavior: There is no significant difference in the typical spending amount between male and female customers.

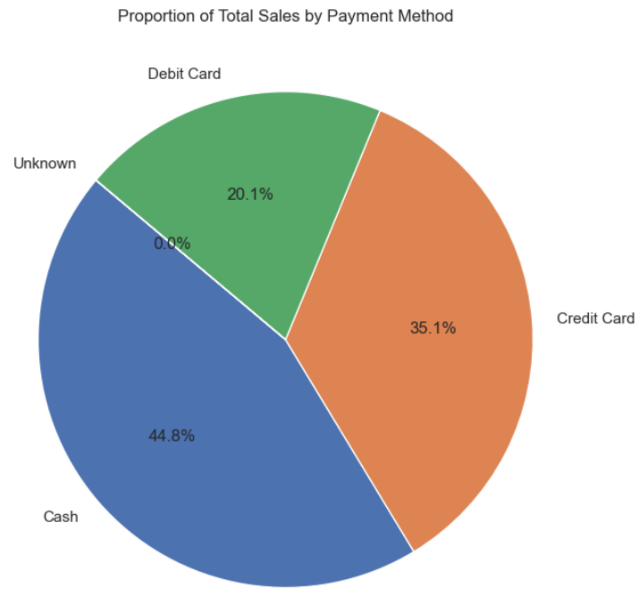


Age

and Spending Correlation: Based on the provided scatter plot, there appears to be a weak positive correlation between age and spending. while there is a slight tendency for spending to increase with age, it's not a consistent or strong trend. Therefore, age should be considered just one of many factors when analyzing or predicting spending behavior.

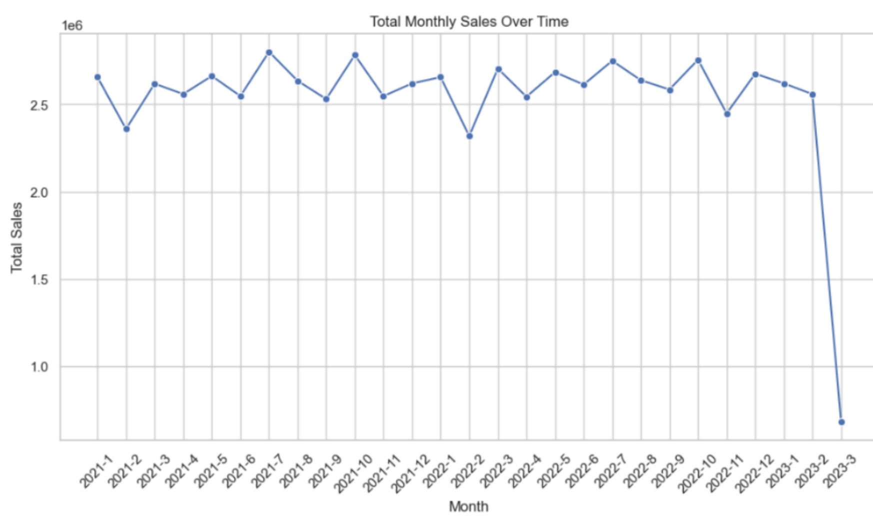


Recommendation:



Payment Methods: Cash is still the top choice for payments, making up 44.8% of transactions, with credit cards close behind at 35.1%. Debit cards also play a significant role, covering 20.1% of sales, and unknown payments are practically nonexistent. This shows that traditional payment methods are still strong, but there's a noticeable shift towards digital payments. To improve, keep cash and credit card options, encourage debit card use with some incentives, look into any unknown payments to fix data issues, and stay on top of new payment trends to keep up with customer preferences.

Sales Trend: Based on the analysis of the line chart for total monthly sales, we observe a steady increase in sales from early 2021 until September 2021, indicating positive growth. Seasonal fluctuations are apparent throughout the year, suggesting that factors like holidays or events influenced sales patterns. However, a significant decline in March 2023 highlights a major disruption that impacted sales. To address these insights, it's crucial to investigate the cause of the 2023 sales drop, identify and leverage seasonal trends for better planning, and develop a strategic growth plan. Additionally, continuous monitoring of key performance indicators will help in adapting to market changes and driving future growth.



Conclusion:

According to the analysis, there is a growing trend towards digital payments, but cash remains the most popular mode of payment. Maintain options for cash and credit cards, incentivize the use of debit cards to increase sales and loyalty, and resolve any discrepancies in the data.

Although sales have typically increased with seasonal variations, more research is necessary in light of a notable decline in March 2023. Make better use of seasonal trends in your planning and create growth-sustaining strategies. In order to succeed in the future and adjust to changes in the market, key metrics should be continuously monitored.

Reference:

1. OpenAI. (2024). *ChatGPT* (Version 4) [Large language model]. OpenAI. https://www.openai.com/chatgpt
2. Google DeepMind. (2024). *Gemini* [AI model]. Google DeepMind. https://gemini.google.com/app